

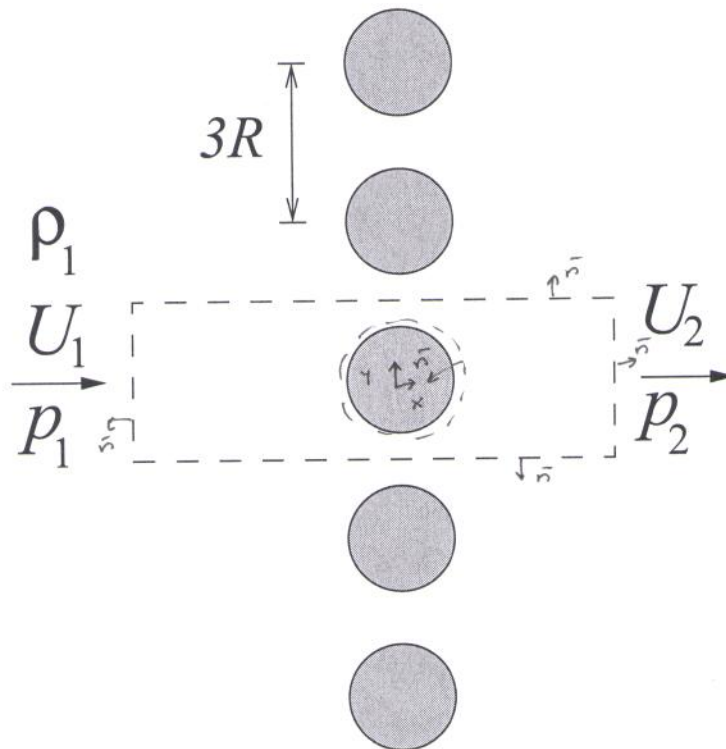
INTRODUCTION TO FLUID MECHANICS

RETAK (Jun 13, 2014)

Problem 1: A linear array of parallel hot cylinders of radius R , whose centers are separated a distance $3R$, is placed perpendicular to a gas stream with velocity U_1 , density ρ_1 , and pressure p_1 , as indicated in the figure. Sufficiently far downstream the gas properties recover uniform values, including a velocity U_2 and a pressure p_2 . Assume that the effect of gravity is negligible. **In terms of the quantities indicated in the figure, determine:**

1. The value of the density downstream from the array of cylinders.
 2. The force acting on each cylinder (per unit length).
 3. The heat transferred from each cylinder to the gas stream per unit time (per unit length).
- Express the result with use made of the ratio of specific heats γ .

NOTE: Explain clearly the simplifications and assumptions employed in the derivation.



CONTINUITY $\int \rho \vec{v} \cdot \vec{n} d\sigma = 0 \quad -\rho_1 U_1 3R + \rho_2 U_2 3R = 0 \rightarrow \boxed{\rho_2 = \frac{U_1}{U_2} \rho_1} \quad (2)$

FORCE $\vec{F} = \int_{\Sigma_{cyl}} p \vec{n} d\sigma + \int_{\Sigma_{cyl}} \vec{\tau} \cdot \vec{n} d\sigma$

MOMENTUM EQ $\left(\int_{\Sigma_c} \rho \vec{v} \vec{v} \cdot \vec{n} d\sigma = - \int_{\Sigma_c} p \vec{n} d\sigma + \int_{\Sigma_c} \vec{\tau} \cdot \vec{n} d\sigma \right)$

$- \rho_1 U_1^2 \vec{e}_x 3R + \rho_2 U_2^2 \vec{e}_x 3R = p_1 \vec{e}_x 3R - p_2 \vec{e}_x 3R - \vec{F}$

$\boxed{\frac{\vec{F}}{3R} = p_1 - p_2 + \rho_1 U_1^2 \left(1 - \frac{U_2}{U_1} \right)} \quad (4)$

ENERGY $\int \rho \left(e + \frac{v^2}{2} \right) \vec{v} \cdot \vec{n} d\sigma = - \int p \vec{v} \cdot \vec{n} d\sigma + \int \vec{v} \cdot \vec{\tau} \cdot \vec{n} d\sigma - \int \vec{q} \cdot \vec{n} d\sigma \rightarrow \boxed{\dot{Q} = \rho_1 U_1 3R \left[\frac{\gamma}{\gamma-1} \left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1} \right) + \frac{1}{2} (U_2^2 - U_1^2) \right]}$

$\frac{U_2}{\rho_1 U_1}$